FACULTY OF COMPUTING

SEMESTER 2 2025/2026

(SECP38843) SPECIAL TOPIC IN DATA ENGINEERING

SECTION 01

GROUP 9

TUTORIAL ARTICLE

LECTURER: DR. ARYATI BINTI BAKRI

STUDENT NAME	MATRIC NO
BRENDAN CHIA YAN FEI	A23CS0211
CHEW CHIU XIAN	A23CS0061
NG YU HIN	A23CS0148

Problem statement

PT. MPM, an Indonesian regional packaging manufacturer, faced a critical gap in its management information infrastructure. The core issues were:

- **Lack of a Performance Monitoring System**
  The organization lacked a suitable application or system for properly assessing and monitoring the performance of its core processes, specifically sales and production. Management had no centralized, real-time view of KPIs.

- **Ineffective Decision-Making**
  Without a proper monitoring system, top management was forced to make business decisions without reliable, timely, or structured data. This reduced the company's competitiveness relative to industry rivals.

- **Fragmented and Unstructured Data**
  Data existed in scattered formats across departments (sales, manufacturing, inventory, warehouse). There was no standardized pipeline to consolidate, clean, and present this data in a usable form.

- **Lack of Production KPI Visibility**
  The production department had no unified view of key indicators such as Overall Equipment Effectiveness (OEE), capacity utilization, quality ratios, and time ratios across machines and production lines.

Proposed solution

The main solution of the problems is **the development of a Self-Service Business Intelligence (BI) Dashboard.** The research proposed developing a BI dashboard solution using Google Data Studio as the visualization platform, with Google Sheets as the data warehouse layer, following the Waterfall SDLC methodology. Two dashboards were delivered:

- **Sales Performance Dashboard**
  Provides senior management with visibility into product sales performance across product categories, sales periods, revenue, forecast vs. actual comparisons, and customer-level breakdowns.

- **Production Performance Dashboard**
  Provides operations management with visibility into OEE metrics (performance/availability/quality), planned vs. actual production counts, and machine-level performance breakdowns via heatmap pivot tables.

Key Technical Components of the Solution:

- ETL pipeline: Raw data → Google Sheets (data cleansing, format transformation, loading) → Google Data Studio
- Data warehouse modelling: Separate sales and production data warehouse schemas
- Interactive controls: Date range filters, product category filters, customer ID filters, machine filters
- Visualization types: Gauge charts, pie charts, combo charts, heatmap tables, bar charts, scorecards
- Access control: Role-based access for Data Analyst and Top Management users

User Stories

The following user stories are derived from the stakeholders identified in the research (Technical & Industrial Design, Inventory & Warehouse, Sales, Manufacturing, Data Analyst, Top Management).

1. Top Management

- As a top manager, I want to view overall sales revenue and compare it against forecasted revenue so that I can assess whether the company is meeting its financial targets.
- As a top manager, I want to view OEE across all production lines in a single interface so that I can identify underperforming machines quickly.
- As a top manager, I want to filter sales data by quarter, customer ID, and product category so that I can drill down into specific performance segments.
- As a top manager, I want to compare planned production vs. actual production so that I can evaluate production efficiency.

2. Operations / Production Manager

- As a production manager, I want to monitor Performance, Availability, and Quality ratios per machine so that I can take corrective action on bottlenecks.
- As a production manager, I want to view machine-level OEE in a heatmap table so that I can rank machines by efficiency at a glance.
- As a production manager, I want to track run time, stop time, and planned production time so that I can identify downtime patterns.

3. Sales Team

- As a sales representative, I want to record and update sales transactions so that the dashboard reflects current sales performance.
- As a sales team member, I want to see product category performance breakdown so that I can understand which product lines are driving revenue.

4. Data Analyst

- As a data analyst, I want to extract, clean, and transform raw reports from ERP into Google Sheets format so that the data warehouse is accurate and up to date.
- As a data analyst, I want to upload processed data to Google Data Studio so that dashboards are refreshed with the latest figures.
- As a data analyst, I want to manage user access to the dashboards so that only authorized personnel can view sensitive data.

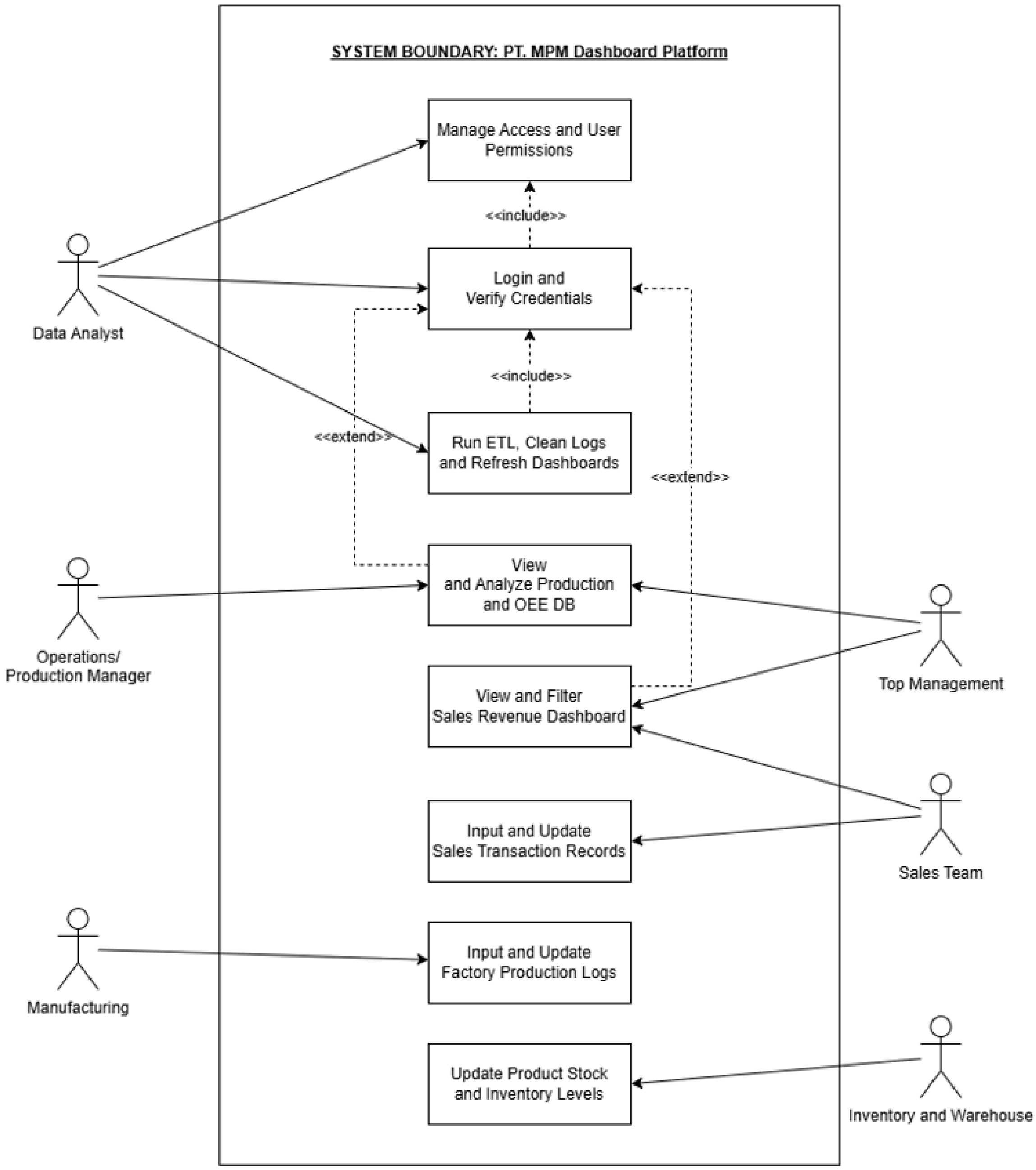
5. Inventory & Warehouse

- As a warehouse manager, I want to update product stock information in the ERP system so that the data used by the dashboard reflects real inventory levels.

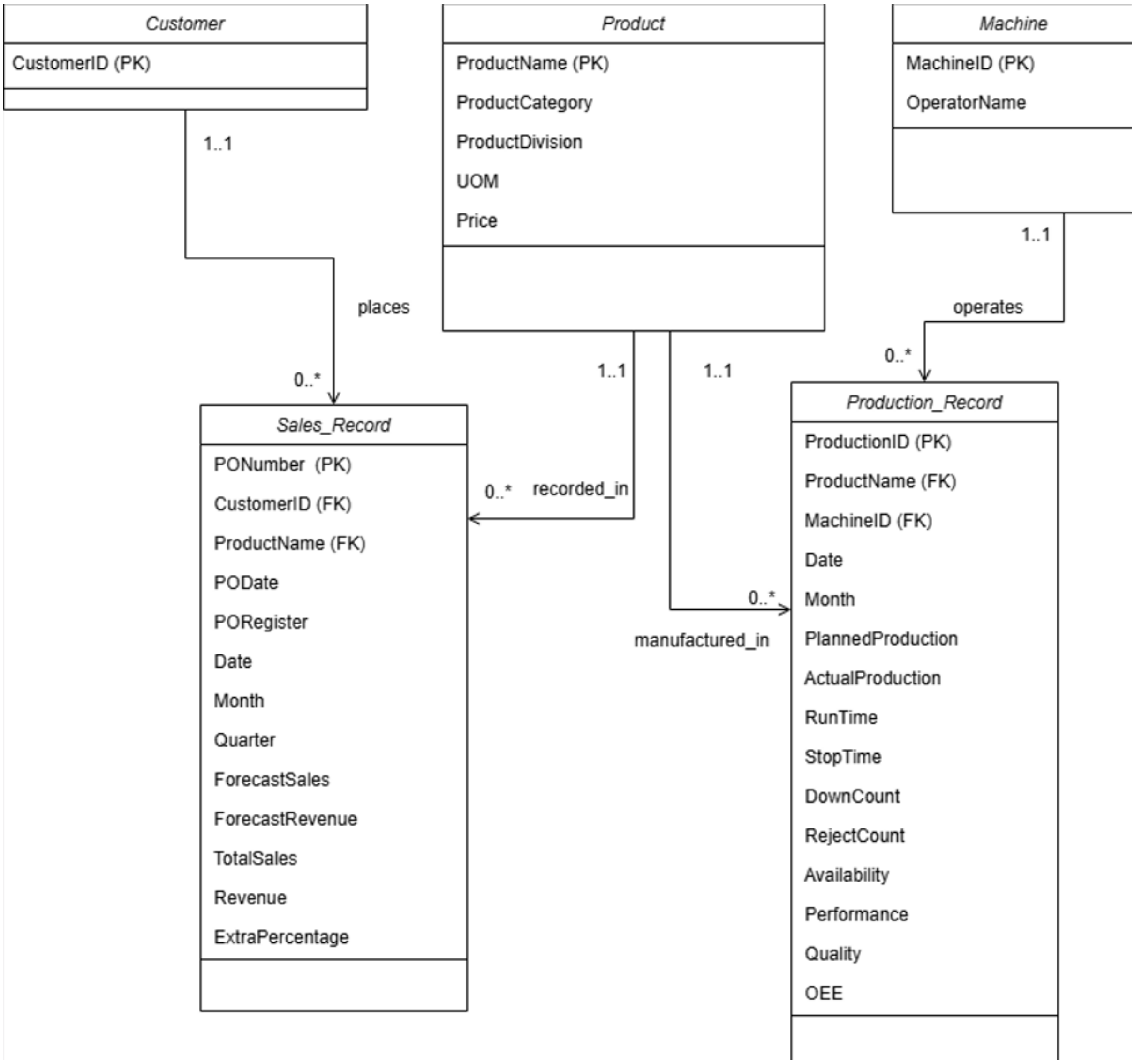
6. Manufacturing

- As a manufacturing worker, I am responsible for tracking operational downtime parameters, parts made, and log updates.

Use Case Diagram



Relational ERD - UML Notation



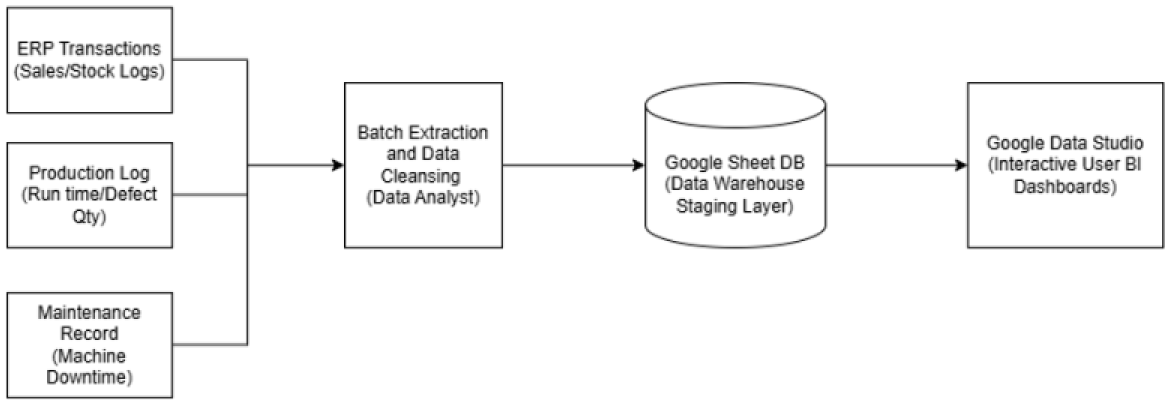
Explanation:

The provided Entity-Relationship Diagram (ERD) illustrates the architectural resolution of a Many-to-Many (M:N) relationship through the implementation of an associative entity. In relational database design, direct Many-to-Many connections between primary dimension tables such as a student registering for multiple classes while a single class accommodates multiple students which creates severe data redundancy and processing anomalies.

To normalize this architecture, the conceptual design is transformed into two distinct One-to-Many (1:N) relationships by introducing a central bridge table, designated in this model as the Enrollment entity. This associative table functions as a transactional record of the interaction between the two primary dimensions. It enforces referential integrity by inheriting the primary keys from both parent tables, utilizing Student_ID and Class_ID as foreign keys to uniquely identify each specific registration event without duplicating descriptive text fields. Furthermore, the Enrollment table serves as the repository for relational attributes, such as the academic grade. The grade attribute cannot logically reside in either the core Student or Class tables independently, as the metric is mutually dependent on the intersection of both entities.

Ultimately, this structural methodology ensures optimal storage efficiency, prevents data duplication, and allows the system to execute complex aggregation queries accurately.

Data & System Architecture



System Architecture Description:

The system architecture follows a linear pipeline that transforms raw operational data into interactive business intelligence outputs across four sequential stages.

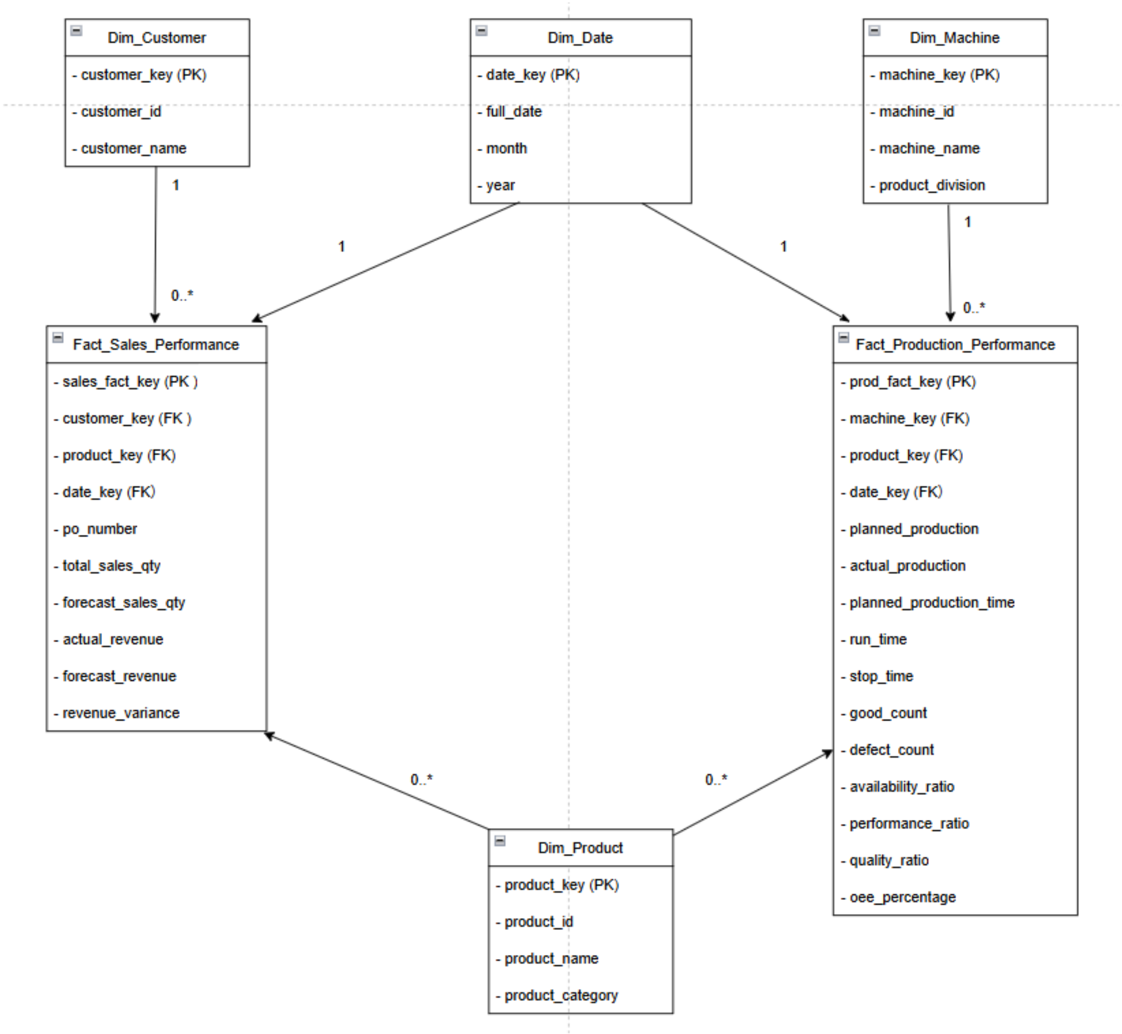
Three distinct source systems feed into the pipeline: ERP transaction records capturing sales and stock logs, production logs recording machine run times and defect quantities, and maintenance records tracking machine downtime events. These heterogeneous data sources represent the operational ground truth of the manufacturing environment.

All three sources converge into a centralised batch extraction and data cleansing process, managed by the data analyst. At this stage, raw records are extracted, inconsistencies are resolved, fields are standardised, and irrelevant or erroneous entries are filtered out before the data advances further downstream.

The cleansed output is then loaded into Google Sheets, which serves as the data warehouse staging layer. Acting as a lightweight but accessible central repository, this layer holds the structured, query-ready data that bridges the transformation process with the reporting front end.

Finally, Google Data Studio connects directly to the staging layer to render the data into interactive user-facing BI dashboards. This front-end layer enables stakeholders including production supervisors and top management to explore sales performance, machine efficiency, and operational trends through dynamic, filterable visualisations without requiring direct access to the underlying data.

Galaxy Schema: Sales & Production Performance Diagram



Structural Attribute List:

- **Fact Table:** Fact_Sales_Performance
  - **Surrogate Primary Key:** sales_fact_key
  - **Foreign Keys (FK):** customer_key, product_key, date_key
  - **Degenerate Dimension:** po_number
  - **Measures (Metrics):** total_sales_qty, forecast_sales_qty, actual_revenue, forecast_revenue, revenue_variance
- **Fact Table:** Fact_Production_Performance
  - **Primary Key:** prod_fact_key
  - **Foreign Keys (FK):** machine_key, product_key, date_key
  - **Measures (Metrics):** planned_production, actual_production, planned_production_time, run_time, stop_time, good_count, defect_count
  - **Calculated Warehouse Metrics:** availability_ratio, performance_ratio, quality_ratio, oee_percentage

- **Dimension Tables:**
  - **Dim_Customer:** customer_key [PK], customer_id, customer_name
  - **Dim_Product:** product_key [PK], product_id, product_name, product_category
  - **Dim_Date:** date_key [PK], full_date, month, quarter, year
  - **Dim_Machine:** machine_key [PK], machine_id, machine_name, product_division

1) Dimensional Data Modeling & Schema Justification

To eliminate slow network speeds and maximize the efficiency of front-end dashboard interactions within Google Data Studio, the third normal form (3NF) core database structure was denormalized into a Galaxy Schema (Fact Constellation) consisting of two interconnected Star Schema Data Marts sharing common conformed dimensions:

- **Sales Performance Mart:** This layout groups data around the central transactional grain found in the `Fact_Sales_Performance` table. It links descriptive dimension models (`Dim_Customer`, `Dim_Product`, and `Dim_Date`) directly via surrogate numerical keys (PK/FK). Quantitative measurement fields, including sales volumes, targets, revenue figures, and calculated variance deltas, are stored on a single flat row layer. This design maps to the top management user stories, allowing stakeholders to filter financial returns across product groups or quarters instantly without running deep multi-table relational joins.

- **Production Performance Mart:** Engineered for factory floor monitoring, the `Fact_Production_Performance` table aggregates primary raw operational values (output volumes, shift running hours, stop timelines, and defective item logs). These base metrics are coupled directly with pre-calculated warehouse processing indicators (Availability, Performance, and Quality components forming the overarching OEE score). Connecting this central hub directly to the flat dimensions of `Dim_Machine`, `Dim_Product`, and `Dim_Date` lets supervisors perform rapid drill-down filtering, isolate equipment slowdowns, and view machine efficiency balances inside dynamic visual heatmaps with zero lag.
- **Conformed Shared Dimensions:** The defining characteristic elevating this design beyond two isolated star schemas into a true Galaxy Schema is the deliberate sharing of `Dim_Product` and `Dim_Date` as conformed dimensions across both fact tables. Rather than maintaining duplicate and potentially inconsistent copies of product and date data, a single authoritative instance of each dimension serves both marts simultaneously. This architectural decision delivers two benefits. First, it guarantees referential consistency, any product category renames or calendar hierarchy change propagates uniformly across both sales and production reporting. Second, it enables cross-mart analysis within a single query, allowing management to directly correlate production OEE scores against sales revenue variances for the same product group and period. Isolated star schemas cannot support this analytical capability without costly ETL reconciliation.